

Could they run AI data centers without using water if there was more room for air flow around computers?

Posted by u/Unusual-Ice-2212 • 0 points • 19 comments

I always hear about how AI uses a lot of water. Why can't they use air cooling, or water cooling with a closed loop (so the water just circulates and doesn't get consumed)? They'd probably have to leave more room for airflow between/around server racks but if the water usage is such a big deal why can't they do that?

Or are people counting the water used by power plants to generate power as "AI water use"?

****edit:** I know using water doesn't delete the water molecules from existence lol. The water turns into vapor and goes into the atmosphere but it was still "used" in the sense that isn't there in whatever lake/river/groundwater source it came from anymore**

Comments

u/isthis_s • 1 points • 2026-04-01 01:12 UTC

why does it have to use fresh drinkable water? Why can't we use impure water or purified water?

u/420FriendlyStranger • 1 points • 2026-02-20 22:24 UTC

Yes, and it has been done. One of the data centers i toured had a Kyoto wheel which offset most of the cooling to airflow.

<https://www2.kyotocooling.com/data-center-cooling-hewlett-packard/>

u/slinkhi • 1 points • 2026-02-20 22:20 UTC

No not really. Not in current state. Water xfers heat faster than air by magnitudes. It would actually be better to go the opposite direction and remove air. As in, put them up into space.

u/GatorBait81 • 1 points • 2026-02-21 05:10 UTC

Space is a vacuum...a wonderful insulator. Not a conductor. Not the opposite direction.

u/UnlamentedLord • 2 points • 2026-02-20 21:33 UTC

They don't cool the chips by spraying water on them lol. There's a loop of coolant that goes to some external radiator outside the building. That's where the water is used.

And yes, the water to generate the electricity is counted in all the data center water usage estimates, so the water usage would actually go up if the data centers used pure air cooling, since that's less efficient and would require more electricity.

u/DiamondJim222 • 2 points • 2026-02-20 21:05 UTC

They do use air cooling. The air is cooled via evaporation in water towers, thus the water use.

u/Unusual-Ice-2212 • 1 points • 2026-02-21 13:18 UTC

What if they just blew the hot air outside and pulled in new cool air instead?

u/makinenxd • 1 points • 2026-02-21 13:45 UTC

You'd need to move the air for such long distances that it would consume way more electricity. Also if you made more room between the servers in a data center you'd obviously have to make the building larger which increases the cost of cabling. When you have already hundreds or thousands of miles of cables its not a small cost if they had to space things more apart.

Water is just a efficient way to transfer heat.

u/MedusasSexyLegHair • 1 points • 2026-02-20 20:00 UTC

They do use both aircooling and closed-loop, but you still need to shed the heat from those systems via immense amounts of HVAC and/or evaporative cooling or something.

Can't really air-cool something sitting in a kiln or cool something by repiping super-hot fluid through it.

u/UnstableUnicorn666 • 0 points • 2026-02-20 18:34 UTC

The AI water usage is way more complex thing.

It's calculated to reach 4-6billion cubic liters next year. For comparison USA corn production, that mostly goes creating ethanol for fuel (not anywhere near best thing to use for ethanol) and to feed farm animals, is around 550billion cubic liters.

And the water used for AI just heats up, small part evaporates, that is still usable water, for as water in farms have pesticides in them. Obviously dumping hot water back to nature has huge effects, that needs to dealt with.

u/StangF150 • 2 points • 2026-02-20 18:29 UTC

OP, water used in cooling turns the water into vapor or steam, it does NOT remove the water from the cosmos. Water vapor then drifts up into the sky becoming clouds, and eventually raining back down to the ground. In other words, all the crap you are seeing low iq people repeat about Ai Data Centers using up water, is bs.

An yes I know I'll get downvoted to oblivion for stating basic science facts, by those that think their feelings overrides the laws of physics.

u/Unusual-Ice-2212 • 1 points • 2026-02-21 14:08 UTC

It's still removing water from whatever source it comes from. Sure it will fall back down as rain but the rain water will fall far away, spread over a large area and won't be easily collected again. Or if the rain falls into the ocean it will just become salt water immediately.

u/slinkhi • 0 points • 2026-02-20 22:21 UTC

Your downvote is not for basic science, but for being a pretentious twat about it.

u/illogictc • 2 points • 2026-02-20 21:54 UTC

The point opponents are making are that it uses large amounts of fresh potable water which is not and cannot just be reintroduced to the system when done. The pumping and sanitizing equipment of a particular town is what it is, if it can do say 300k gallons per hour then if the usage already nears that threshold, what will a data center which needs large quantities of water mean for that capacity? Unless the data center will also be making an investment in expanding the town's potable water capacity, and also assuming that expansion isn't going to mess too badly with those on well water etc because there are also water tables to consider and all that.

It just needs to be planned with this stuff in mind and a realistic and honest look at **all** of the effects both positive and negative the center means not just for the company but also the community.

u/adsarelies • 1 points • 2026-02-21 08:46 UTC

Aren't those questions something the utility company should be thinking about, with the money they get from the massive water bills?

u/illogictc • 1 points • 2026-02-21 09:06 UTC

They won't be getting that money until after it's built, and there's also a question about any sort of bulk pricing they may negotiate if that's a possibility. Then you can't just plop something down from Sunbelt Rentals within a couple hours, it takes time to design, build, and commission. And in many localities, the local water system is owned by the community itself; it's the town's water supply, it's not some big megacorp with billions in the bank. About 7/8 of all water systems are owned and operated by the town/city.

u/adsarelies • 1 points • 2026-02-21 09:31 UTC

OK, maybe the data center owners should think about where they build, perhaps not in areas served by small time utilities.

But I feel it's kinda unfair to expect a utility customer to invest in the local utility infrastructure, on top of the bills they pay. Is there precedence for this? Like, do the farms front money to local utilities for using water? I don't know, not a rhetorical question. And if they do, does that mean now they own part of the local utility? Is that something we want?

u/illogictc • 1 points • 2026-02-21 12:46 UTC

Farms are essential. Nobody can argue they're not unless they wanna just not eat to prove how non-essential they are. Is an AI data center really that essential? So far AI seems to be mostly used as a really advanced

chatbot, a research tool that can be used correctly or incorrectly (where hallucinations become part of the final work), and for making images and videos for purely entertainment or misinformation purposes. Are these essential services? And is it worth [just how thirsty these things are?]

(<https://www.eesi.org/articles/view/data-centers-and-water-consumption>) perhaps in the future they'll prove themselves a lot more handy than they are now. Also note this article cites sources that also raise the problem of stressed watersheds and that it's not talking AI data centers in particular, which are even more resource-hungry.

I just mention investment in the local provider as a potential avenue to offset the fact they they're going to soak up hundreds of thousands or even millions of gallons of water per day, and then are going to evaporate a large portion of it, which scatters it to the four winds. Or they can buy their own infrastructure which some data centers already do, tapping directly into aquifers etc and doing whatever purification makes it good enough to make a run through their cooling cycle even if it's not necessarily considered good enough to call potable. This is something other companies already do, like a hospital purchasing its own electric generation infrastructure to act as a backup.

u/Nuts4WrestlingButts • 5 points • 2026-02-20 18:50 UTC

Nobody is saying that cooling systems use up the water and remove it from the cosmos. Fresh, potable water is what's getting used. Fresh water is a limited resource that takes a lot of time and energy to produce. It turns into vapor and then rains down hundreds of miles from where you took it.

u/generic_redditor_71 • 8 points • 2026-02-20 18:02 UTC

They could, but it would be less energy-efficient, which is actually the important factor for costs and environmental impact of data centers, not their negligible water consumption.